CS484 Project Progress Report: Landmark Classification Using Resnet-18 Cifar Model

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I. Introduction

Land-use classification from satellite imagery is an important task in remote sensing, which supports applications such as environmental monitoring and urban analysis. Due to the visual similarity between classes and the low resolution of the images, this task necessitates models that can extract fine-grained spatial features.

In this project, we used the ResNet-18 Cifar model to classify the ten EuroSAT land-use categories. The model's residual connections and lightweight architecture make it suitable for small-scale image recognition. We train and evaluate the model under different hyperparameter settings to analyze its convergence behavior and overall classification performance. The results demonstrate the effectiveness of residual networks for satellite image recognition, highlighting the impact of training duration and learning rate on model accuracy.

II. Problem definition

Our goal in this project is to develop a deep learning model that accurately classifies ten land-use categories from satellite imagery using the EuroSAT dataset. Satellite images present unique challenges due to their low resolution, high intra-class variability, inter-class similarity, and atmospheric distortions. To classify satellite images, we require a model that can extract fine-grained spatial features while maintaining robustness against noise and subtle texture differences. As a solution, the ResNet-18 Cifar model is selected for fine-tuning. ResNet-18 Cifar is specifically adapted for small images, enabling the model to preserve spatial detail and learn discriminative features effectively.

III. Dataset

The EuroSAT dataset is a popular remote sensing benchmark. It consists of 27,000 RGB images (64 x 64 pixels) from Sentinel-2 satellite imagery. It contains around 10 balanced land-use categories, including forest, residential, sea/lake, and agricultural areas. Due to its diversity, EuroSAT is a key tool for evaluating and comparing deep learning models in satellite image recognition and classification [1].

IV. Model architecture

ResNet-18, introduced initially by He et al., is a convolutional neural network that uses residual learning to make deeper models trainable and more accurate [2]. Instead of learning a direct mapping, each block learns a residual function that is added back to the input via a skip connection. In the project, a CIFAR-style variant of ResNet-18 is employed, specifically adapted for small-resolution images by changing the stem layer and preserving higher spatial detail in the early stages of the network. The architecture can be expressed as a set of convolutional, normalization, activation, pooling, and fully connected operations organized in a residual hierarchy that enables efficient feature extraction, stable optimization, and strong generalization performance across diverse visual datasets.

The network begins with a simplified stem consisting of a single 3x3 convolutional layer, instead of a 7x7 convolution or max-pooling layer, as in the ImageNet-scale ResNet. This 3x3 convolution layer maps the input RGB image into 64 feature channels, followed by batch normalization and ReLU activation, which prevents overly reducing the spatial resolution of small images. Instead, the architecture of Cifar-style ResNet preserves fine-grained information early in the forward pass to support more effective learning on compact datasets. After the stem, the architecture transitions into a series of four primary residual stages that expand the feature dimensionality to 64, 128, 256, and 512 channels while progressively decreasing spatial resolution through strided convolutions.

Each stage is composed of two BasicBlocks, and each BasicBlock contains two consecutive 3x3 convolutional layers, both followed by batch normalization, with a ReLU applied after the first convolution. The main feature of the ResNet family is its skipping connections, which add the input of the block directly to its output. This procedure helps to train deeper networks without suffering from vanishing gradients. When the spatial resolution or number of channels changes between

blocks (as is the case at the start of stages 2, 3, and 4), a 1x1 convolution with a stride of two is applied on the shortcut path to ensure compatibility with the main branch. This configuration allows the network to deepen its representational capacity while maintaining stable optimization behavior.

As the model progresses through its four stages, the first block in each stage employs a stride of 2 to downsample the feature maps, thereby increasing the spatial context available to the network to recognize higher-level and more abstract patterns. The remaining blocks in each stage use a stride of 1, preserving resolution and focusing on deepening the feature representation through additional nonlinear transformations. This gradual combination of downsampling and residual learning results in a balanced architecture that can capture both local texture patterns and global contextual information.

After passing through the final residual stage, the network applies an adaptive average pooling operation that reduces each feature map to a single scalar, resulting in a fixed-length feature vector of size 512, for all input dimensions. This vector is then passed through a dropout layer, which provides regularization by preventing overfitting, and finally through a fully connected linear layer that maps the learned representation to the ten classes defined in the EuroSAT dataset. These final-layer activations correspond to the model's estimates for each class. The complete architecture reflects a streamlined and computationally efficient residual network design, explicitly optimized for small images.

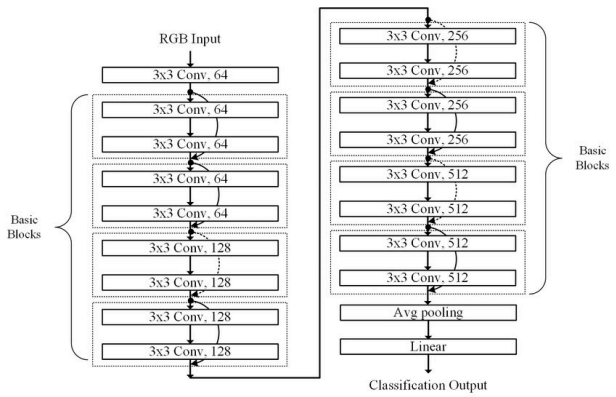

Fig. 3. Architecture of Resnet-18 Cifar Model

V. Training and Results

The EuroSAT dataset was utilized for all experiments. To guarantee a balanced evaluation and reliable generalization, the dataset was systematically partitioned into distinct sets: 80% for training, 10% for validation, and 10% for testing.

Training run #1 produces the most successful outcome, achieving a final accuracy of 96.19%. The configuration was as follows:

- Epochs: 10
- Batch size: 64
- Learning rate: 1×10^{-3}
- Weight decay: 1×10^{-4}
- Dropout: 0.2
- Width multiplier: 1.0

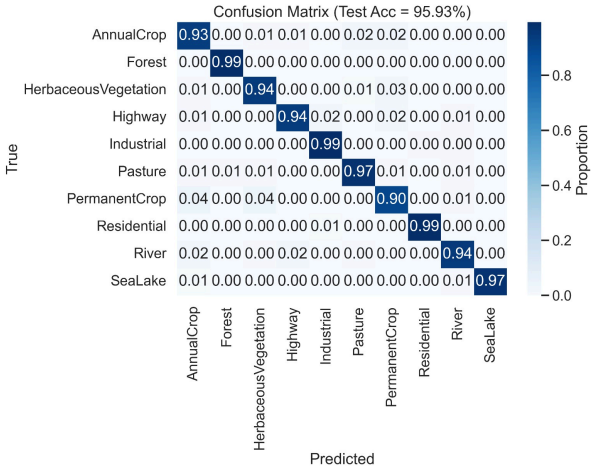

Fig. 4. Confusion matrix of the results on the test set

Fig. 5. Performance of Resnet-18 Cifar on various metrics

Both the loss and accuracy curves show smooth convergence, which indicates that the learning rate and weight decay parameters were well balanced for this dataset.

Validation metrics such as Precision, Recall, and F1 score also stabilized after the sixth epoch, suggesting that ten epochs were sufficient for full convergence.

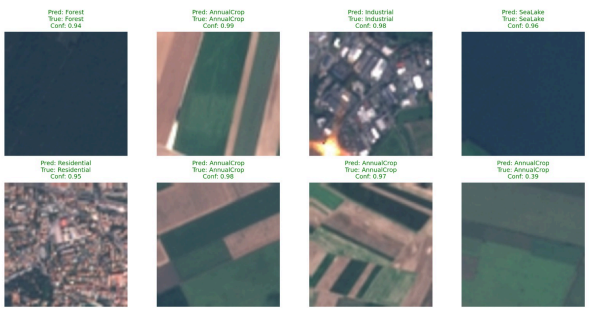

Fig. 6. Land classification performance of the 1st training

Effect of Epoch Number

Training #2 was executed with the identical hyperparameters as Training #1, but was limited to 5 epochs to assess its effect on the training. This configuration produces a final accuracy of 92.41%, which was achieved at epoch 5. Although the model's performance remains strong, it did not match the accuracy of the first training, which illustrates that this learning rate (1×10^{-3}) requires a longer training time for the model to fully converge.

Effect of Learning Rate (LR)

In the third run, the model was trained for 10 epochs with a learning rate increased to 1×10^{-1}, while all other parameters remained unchanged. This higher learning rate resulted in unstable weight updates and fluctuating loss values, which resulted in a lower accuracy of 91.41%. We observed that excessively large learning rates can prevent convergence and negatively affect model generalization.

REFERENCES

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[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 770–778, 2016.